Region-Growing-Based Automatic Localized Adaptive Thresholding Algorithm for Water Extraction Using Sentinel-2 MSI Imagery

Bharath Kumar Reddy Kadapala and Abdul Hakeem K

***Abstract*— Water is a distinct land cover feature on the Earth. Water can easily be extracted from satellite data under known/controlled conditions using simple thresholding techniques. However, these pixel thresholding techniques fail when applied over large regions because they cannot adapt the thresholds based on local variability. Although region-growing methods are available to overcome this issue, there is a limitation in using a global threshold value determined for a satellite scene. Such thresholds fail to adapt to the variations within the scene. Hence, this study proposed a new region-growing-based approach for improved classification of water features using Sentinel-2 multispectral instrument (MSI) data. The significant difference between this approach and other region-growing methods is that this technique uses dynamic thresholding for localized adaption. The local image statistics of the normalized difference water index (NDWI) layer are used for adding new pixels to the region, and the threshold is adjusted for every new pixel added. This technique also minimizes manual intervention in water classification, thus enabling total process automation in near real time. The thresholds are dynamically determined not only for each scene but also for each new pixel being classified for the accurate delineation of the water pixels. A rigorous quality evaluation was done on the resultant water layer. This technique achieved an overall accuracy of 84.04% in challenging conditions.**

***Index Terms*— Adaptive thresholding, normalized difference water index (NDWI), region growing, water monitoring, water spread area.**

I. Introduction

INLAND water bodies affect directly or indirectly most of flora and fauna around them. Better management of water bodies is not only essential to the survival and sustenance of its dependent sectors but also essential to maintain balance in its ecosystem. Quantity and quality are regarded as important indicators of water security in a region [1]. Inland water spread area is one of the important and direct indicators of the quantity of water available in many regions and hence also for water security. Because of the nonperennial nature of the major rivers in southern and central India, water bodies are major sources of water in the region during the non-monsoon period. Making an inventory and monitoring the water spread area of water bodies is the first step toward better management of the water bodies and addressing the water security of this region.

Sentinel-2 mission by the European Space Agency (ESA) provides an opportunity of analyzing the Earth's surface at an unprecedented temporal resolution (five days) using a multispectral instrument (MSI) at a reasonably high spatial resolution of 10 m. Many researchers worldwide use these data to generate multiple information products and extract various features [3], [4], [5], [6], including water. Considering many of the spectral bands in Sentinel-2A/B similar to that of Landsat-8, many techniques and indices employed with Landsat-8 for water extraction can also be used for Sentinel-2. The higher resolution of the sensors further improves the accuracy of the water spread area estimated using Landsat-8 [6]. This study was carried out to develop an automated water extraction technique using the high spatial and temporal resolution of Sentinel-2A/B to monitor the surface water dynamics.

Although spectral characteristics of water are unique compared to other dominant land cover features, mapping and extraction of inland water bodies, especially urban water bodies, is considered a challenging task. This is due to the difficulty in eliminating regions with topographic shadows, cloud shadows, built-up areas, snow, and ice [7]. It is a known fact that many indices, such as normalized difference water index (NDWI) [8], Modified NDWI (MNDWI) [9], automated water extraction index (AWEI) [10], and water index (WI) [11], are developed to accurately delineate water, accounting some of the earlier mentioned challenges. Studies were carried out earlier for the water delineation through a knowledge-based algorithm using multitemporal spectral bands of the advanced wide field sensor (AWiFS) of Resourcesat-1 [14]. Image sharpening approach for urban surface water extraction [15] and adaptive remote sensing extraction algorithm framework for the complex muddy coast waterline using Sentinel-2 MSI data [16] were also used for water delineation. Pixel unmixing techniques were also employed to improve the accuracy further. Thresholding is one of the simplest and most widely used techniques. However, determining thresholds remains a critical issue in water extraction and is found to produce better results if determined manually [2]. To account for this problem, some researchers worked on developing dynamic and adaptive thresholding when mapping large regions [12], [13].

Many adaptive thresholding techniques use a fixed threshold for a smaller region or a subscene of the larger image [17],

Manuscript received 16 December 2022; accepted 14 February 2023. Date of publication 20 February 2023; date of current version 1 March 2023. *(Corresponding author: Bharath Kumar Reddy Kadapala.)*

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Digital Object Identifier 10.1109/TGRS.2023.3246540

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TABLE I
CHARACTERISTICS OF SENTINEL-2A BANDS

Bands	Central wavelength (nm)	Spatial resolution (m)
1 - Coastal aerosol	442.7	60
2 - Blue	492.4	10
3 - Green	559.8	10
4 - Red	664.6	10
5 - Vegetation red edge	704.1	20
6 - Vegetation red edge	740.5	20
7 - Vegetation red edge	782.8	20
8 - Near Infrared (NIR)	832.8	10
8A - Narrow NIR	864.7	20
9 - Water vapour	945.1	60
10 - SWIR - Cirrus	1373.5	60
11 - SWIR-1	1613.7	20
12 - SWIR-2	2202.4	20

[18], [19]. This same local threshold determined is used for the entire subscene or local region. If the intrascene variability in water pixel characteristics is high, then these techniques also suffer similar disadvantages as that of the global thresholding methods. This study aims to develop a region-growing technique with localized adaptive thresholds to accurately extract water spread by differentiating water from other features.

II. METHODOLOGY

A. Objective

This study aims to develop an automatic water extraction algorithm with localized adaptive thresholds. In contrast to other techniques which use the same local threshold based on the image statistics of the local region, the threshold/condition for water delineation (growing the region) is required to adapt and change with the classification of every new water pixel.

B. Data and Preprocessing

The proposed algorithm was developed for imagery from the MSI of Sentinel-2A/B. The Level-2A bottom-of-atmosphere corrected reflectance products distributed by ESA via Copernicus Open Access Hub were used in this study. The characteristics of the spectral bands available from MSI of Sentinel-2A are given in Table I. The Sentinel 2B bands also have almost identical sensor characteristics, and differences can be safely ignored for this study.

In this study, green (Band-3), red (Band-4), near-infrared (NIR) (Band-8), and short-wave infrared (SWIR-2) (Band-12) bands of Level-2A products were used for generating the water layer. The Level-2A product also provides quality indicator bands that include a cloud probability layer. This layer was also utilized to remove clouds and the surrounding regions affected by its shadow. Green and NIR bands were used to derive the NDWI [8], while Green and SWIR-2 bands were used to derive the modified NDWI-2 (MNDWI-2), an index similar to MNDWI [9]. Green, red, and NIR bands' reflectance data were also used along with the others in filtering the clouds in addition to the cloud probability layer. The overview of the algorithm is depicted as a flowchart in Fig. 1.

Cloud masks were prepared by reclassifying the cloud probability layer into a binary image with a probability greater than 50%. Morphological erosion was performed on the binary cloud mask images to eliminate small parcels of misclassified areas. Morphological dilation was then performed to eliminate the shadow regions around the clouds. A square structuring element of 5 pixels was used as an erosion element, and 100 pixels were used as a dilation element. In addition, manual thresholds over the average reflectance in green, red, and NIR bands were used to improve the mask layer. These cloud masks were used in the latter part of the processing.

C. Preparation of Water Masks

The detection of surface water pixels is divided into two steps in this study. The first step is identifying high-confidence water pixels with known spectral characteristics, usually unmixed with other features. With the help of indices mentioned in the latter part of this section, some of the mixed pixels with relatively lower content of land characteristics can also be classified with high confidence. All the pixels with the above characteristics are called high-confidence pixels in this study. With prior knowledge of the spectral properties of water under known conditions, a fixed global threshold technique is used to determine the high-confidence pixels.

The second step identifies the low-confidence pixels, usually mixed/impure, with a higher percentage of land characteristics. In this study, the pure water pixels affected by sun glint, with high turbidity or suspended sediments, are also called low-confidence pixels.

Modifying the threshold on the water indices to include all the low-confidence pixels in a region might result in the addition of unwanted features into the classification in another region. Hence, fixed threshold-based techniques might fail to deliver the accuracies required when operating on larger and nonhomogenous datasets. Some studies reported a difference of 20% in the accuracy between identifying pure and mixed pixels [11]. Hence, in this study, these fixed threshold-based techniques were used to classify only high-confidence pixels. A threshold value of NDWI $>$ 0 in noncloudy regions can be classified as a highly confident water pixel. In addition to the criterion mentioned above, the pixels with an average reflectance of NIR and SWIR2 less than 0.1 and the average of NDWI and MNDWI-2 greater than 0 were also added to high-confidence pixels. These pixels act as seeds for growing the region.

A mask of low-confidence pixels was also prepared with relaxed thresholds, with average reflectance in NIR and SWIR-2 less than 0.15 and the average of NDWI and MNDWI-2 greater than –0.2. All these pixels might not necessarily be water but would provide a valuable buffer around the identified seed pixels into which they can be grown. All the conditions applied while generating the low-confidence mask can also be applied while growing the region. This

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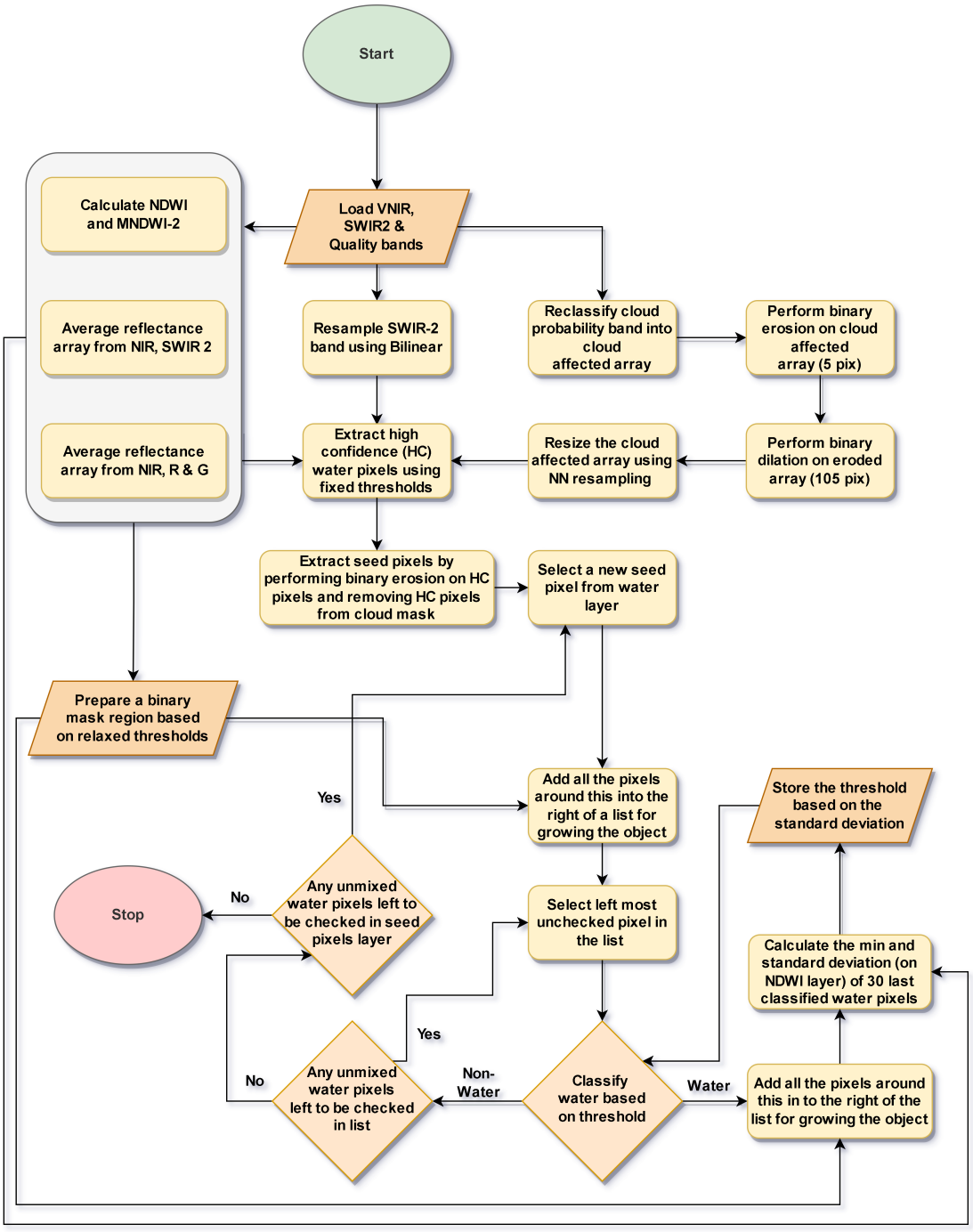

Fig. 1. Methodology overview of the algorithm.

mask was made only to improve the computational efficiency of the algorithm. Preparing the low-confidence mask prior would decrease the computational and memory needs of the algorithm. The cloud mask was used at this stage to modify the low-confidence mask by removing the pixels inside the mask.

Most of the high-confidence water pixels would be used in further steps as seed pixels for region growing. Morphological erosion was performed over these high-confidence pixels to erode the noise or small individual pixels. Furthermore, high-confidence pixels inside the dilated cloud mask were omitted, and the remaining pixels were used as seed pixels.

D. Region Growing

The region-growing technique was initially applied over an image/layer generated by averaging NDWI and MNDWI-2 layers. Since NDWI is very sensitive to nonwater features, the MNDWI-2 layer was added and the averaged layer (of NDWI and MNDWI-2) was planned to be used. However, since moist/damp soils and water with low/sparsely spaced floating aquatic plants have similar spectral reflectance, regions

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growing over this layer introduced significant commission errors. Especially during the post-monsoon period, the commission errors were found to be higher in the plains of North India and peninsular India, where wet and moist soil exists. Hence, the region growing was finally performed over the NDWI image instead of the averaged image. This reduced the commission errors in almost all the regions in pre- and post-monsoon periods, with the effect predominantly visible in the post-monsoon season.

The high-confidence pixels and low-confidence masks were used along with this NDWI image to perform the region growing. Seeds were identified, and the growing begins with the selection of the first seed. The criterion to include a new pixel from the NDWI raster into the region was based on the minimum and standard deviation of a fixed number (30 pixels) of local and previously identified/classified pixels (30 pixels). This helps to estimate the thresholds locally and becomes adaptive to the change in the region and characteristics of water bodies.

The initial conditions of minimum (min) and standard deviation (of NDWI and MNDWI-2 averaged raster) are configurable and are not too sensitive since they change immediately to get adapted to locally identified water pixels. The initial conditions used in this algorithm were min –0.05 and standard deviation 0.05. The lower bound of the threshold (Th_{lb}) was modified after every new pixel was added into the region. The value of Th_{lb} was determined as given in the following:

$$\mathrm{Th}_{lb} = \mathrm{Min}(x_i) – n * \mathrm{Std}(x_i) \tag{1}$$

where x_i represents the fixed number of pixels recently classified as water, n is a configurable scaling factor for which the default value is 1, and $\mathrm{Std}(x_i)$ is the standard deviation of the sample.

The parameter "n" can be configured by increasing/decreasing its value, which facilitates the inclusion and exclusion of boundary pixels with varied proportions of the spectral contribution of water. A higher value of "n" accepts a pixel with a smaller proportion of water, while a smaller value of "n" helps eliminate the boundary pixels. The total number of pixels used to calculate mean and standard deviation is also configurable based on the requirements. A smaller group of pixels allows for larger variability while growing the region. Hence, it is suitable for regions with high variability in spectral characteristics of water. A relatively homogeneous region is unaffected by the change in this parameter.

Once a new pixel is added, the neighbouring pixels will be examined to include in the region. This region grows until it cannot expand further due to the lack of suitable pixels that can be added. The algorithm then searches for another seed to grow. The algorithm finishes processing once all the seeds and their respective regions are analyzed.

E. Accuracy Assessment of Algorithm

Multiple regions across India were selected for assessment of this algorithm's efficiency in the extraction of water. A total of 14 image scenes were identified across India from the following regions.

1) North-Eastern India.
2) Himalayan region.
3) North Indian plains.
4) Western Ghats.
5) Southern peninsular region.
6) Central-Western.
7) Western Arid/Semi-Arid.

One scene each from the above regions was selected in the post-monsoon period (October–December), while another scene from all the regions was selected in the pre-monsoon period (Mid-February to Mid-May).

Metrics, such as total accuracy of the classification, producer's accuracy, and user's accuracy of water class, were used to analyze the results.

1) Validation Procedure: Since the preparation of reference masks for all the scenes is error-prone, a systematic cluster sampling approach was used to estimate the algorithm's accuracy.

In pixel-level accuracy assessment, clusters of pixels were identified with 100 pixels in each cluster. Ten such clusters were selected for each scene, with each cluster among the three categories, namely, water, nonwater, and water and nonwater boundary. At least three clusters from each category were selected for each scene randomly. The individual pixels of various categories in all the clusters were clumped in each scene, and accuracy statistics were generated.

If T_w is the total number of water pixels present in raster, t_w is the total number of pixels classified as water and $(T_w \cap t_w)$ represents the total pixels classified as water in both reference and algorithm output. The subscript "nw" represents nonwater counterparts of the above.

User's Accuracy (water) $= (T_w \cap t_w) * 100/t_w$

Pixel level Commission Error of water class (PCE) = 100-User's Accuracy

Similarly,

Producer's Accuracy (water) $= (T_w \cap t_w) * 100/T_w$

Pixel level Omission Error of water class (POE) = 100-Producer's Accuracy

Total Pixel Level Accuracy (TPLA) $= [(T_w \cap t_w) + (T_{nw} \cap t_{nw})] * 100/(T_w + T_{nw})$.

III. Results and Discussion

A. Overview of Results

Based on the validation procedure mentioned above, 14 scenes were selected from different regions across India for the accuracy assessment. The accuracy across the regions varied from 76.15% in the Himalayan region to 93.85% in the Western Ghats region. The overall total accuracy assessment at the pixel level is observed as 84.04% for all 14 scenes. The overall total accuracy assessment at the pixel level is observed as 79.87% and 88.2% for pre-monsoon and post-monsoon seasons, respectively. Tables II–V provide the summary of the accuracy assessment results overall and seasonwise.

The difference between the user's accuracy in pre- and post-monsoon is less than 1%. This indicates that the misclassification in the nonwater pixels is consistent in both seasons. However, a significant difference of 15.41% between pre- and post-monsoon seasons is found in the producer's accuracy.

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TABLE II
OVERALL ACCURACY METRICS

S. No.	Metric	Percentage
1	User's accuracy for water class	92.96
2	Commission error for water class	7.04
3	Producer's accuracy for water class	76.29
4	Omission error for water class	23.71
5	Total accuracy	84.04

TABLE III
POST-MONSOON ASSESSMENT OVERALL ACCURACY METRICS

S. No.	Metric	Percentage
1	User's accuracy for water class	93.26
2	Commission error for water class	6.74
3	Producer's accuracy for water class	84.07
4	Omission error for water class	15.93
5	Total accuracy	88.20

TABLE IV
PRE-MONSOON ASSESSMENT OVERALL ACCURACY METRICS

S. No.	Metric	Percentage
1	User's accuracy for water class	92.60
2	Commission error for water class	7.40
3	Producer's accuracy for water class	68.65
4	Omission error for water class	31.35
5	Total accuracy	79.87

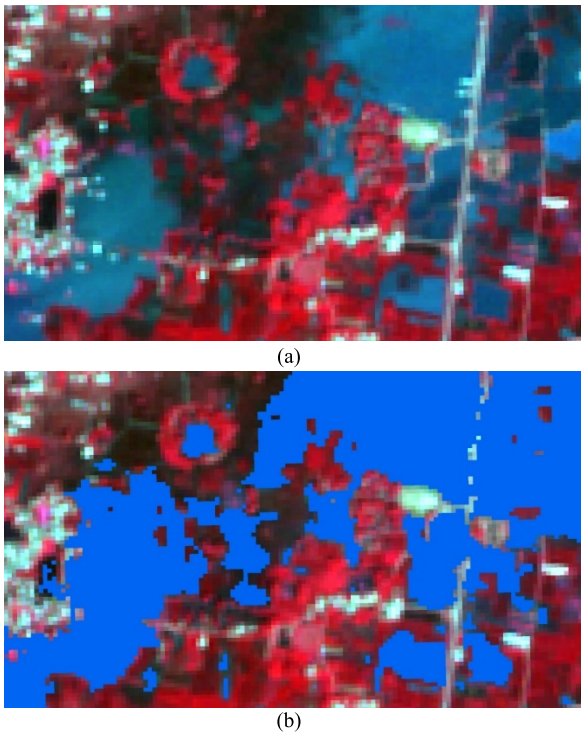

Fig. 2. Water layer extraction for flooded areas in North Indian plain. (a) False-color composite (FCC) image. (b) Extracted water layer overlaid on FCC image.

In pre-monsoon, most of the waterbodies are generally at their lowest water spread area. In regions dominated by small water bodies (<1 ha), the binary erosion operation performed over the high-confidence pixels resulted in reduced seed pixels in such areas. Due to this, region growing could not be initiated and resulted in the omission of many water pixels, especially on data from North Indian plains and Southern peninsular regions. Hence, the post-monsoon producer's accuracy was better in these regions as the seed pixels for region growing were available. In particular, this algorithm performed well in flooded areas (Fig. 2), where global thresholding typically fails due to the boundary and mixed pixel problem.

The scenes from the Central-Western and Western Arid regions of India had small waterbodies dried up completely in the pre-monsoon. In post-monsoon, the water spread area increased, and water was found in smaller waterbodies too, including many farm ponds which were dry/had less water in pre-monsoon (Fig. 3). Although the high-confidence pixels were identified in some of these waterbodies, the binary erosion operation removed them as the seed pixels. This resulted in a decrease in producer accuracy in this region in the post-monsoon season.

An assessment of the algorithm performance was done by combining both the seasons and the accuracy estimated is given in Table VI. It is observed that the algorithm performed best in the Western Ghats, which is hilly terrain, with an accuracy of over 93%. However, in the Himalayan region, the accuracy was lowest, at around 76.15%. Although the terrain was rugged in both cases, the steep slopes and less vegetation made the classification in the Himalayan region difficult.

In addition to these factors, the snow deposition (Fig. 4) over the thin ice layer formed over the waterbodies made the classification even more challenging. Despite these issues, the algorithm performed well in most waterbodies mixed with a thin ice layer. Although the ice layer is formed over the water, the region growing performed satisfactorily in these areas achieving over 76.15% accuracy. These results indicate the effectiveness of the algorithm in tough scenarios where the traditional global thresholding approaches struggle to adapt to local conditions.

B. Advantages and Shortcomings

Since this algorithm modifies the thresholds based on local conditions, it is observed that if the spectral signature of water and mixed pixels changes gradually, the classification performed exceptionally well across the regions. It is also observed that the algorithm performed well across waterbodies of different sizes, shapes, and conditions. The adaptive nature of this algorithm was tested in various challenging scenarios and the performance was found satisfactory in most cases.

However, the difficulty in classifying mixed water pixels was also observed if the spectral characteristics of the mixed pixels are significantly different from water (Fig. 5). In Fig. 5, the sparsely spaced floating vegetation where the change in NDWI is lower and gradual is classified as water, while the densely spaced floating vegetation could not be classified as water due to the abrupt change in NDWI values. Since region growing is performed on the NDWI layer, mixed pixels with

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TABLE V
SEASON WISE PRODUCERS AND USERS ACCURACY FOR DIFFERENCE REGIONS

Region	Pre-monsoon			Post-Monsoon		
	Users Accuracy	Producers Accuracy	Total accuracy	Users Accuracy	Producers Accuracy	Total accuracy
North-eastern	96.51	69.50	82.90	97.74	78.90	86.20
Himalayan region	85.77	68.21	70.30	79.67	92.37	82.00
North Indian plains	99.27	47.48	70.30	93.83	94.61	94.40
Western Ghats	95.77	69.62	90.20	98.28	96.28	97.50
Southern peninsular	96.85	56.68	70.60	94.16	89.66	91.70
Central-Western	94.39	80.04	87.00	97.17	66.64	81.60
Western Arid/Semi-Arid	88.21	91.24	87.80	99.19	70.04	84.00

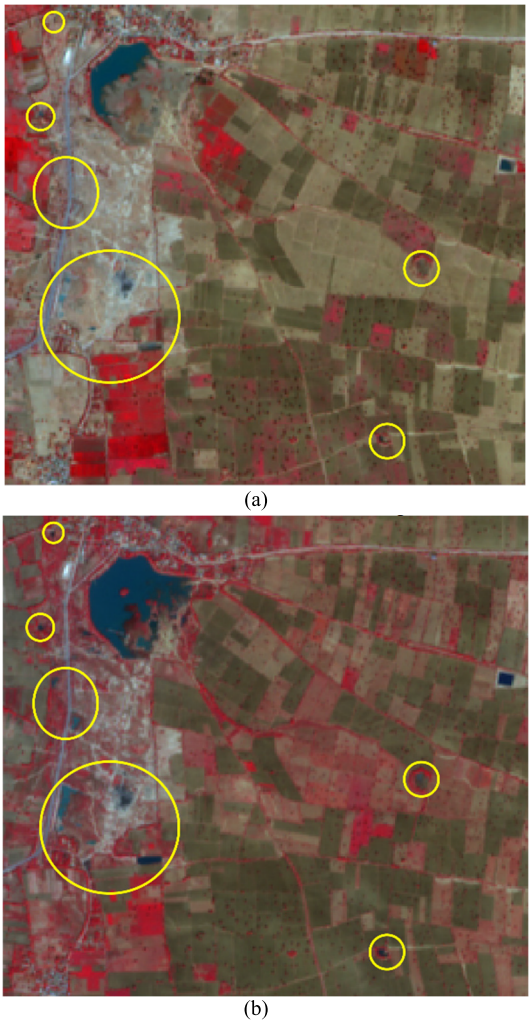

Fig. 3. Pre- and post-monsoon images in the Central-Western region of India. (a) Pre-monsoon FCC image. (b) Post-monsoon FCC image.

TABLE VI
REGION WISE PRODUCERS AND USERS ACCURACY

Region	Users Accuracy	Producers Accuracy	Total accuracy
North-eastern	97.21	74.55	84.55
Himalayan region	82.42	72.36	76.15
North Indian plains	95.78	69.37	82.35
Western Ghats	97.55	87.23	93.85
Southern peninsular	95.33	71.36	81.15
Central-Western	95.62	73.25	84.30
Western Arid/Semi-Arid	92.39	81.19	85.90

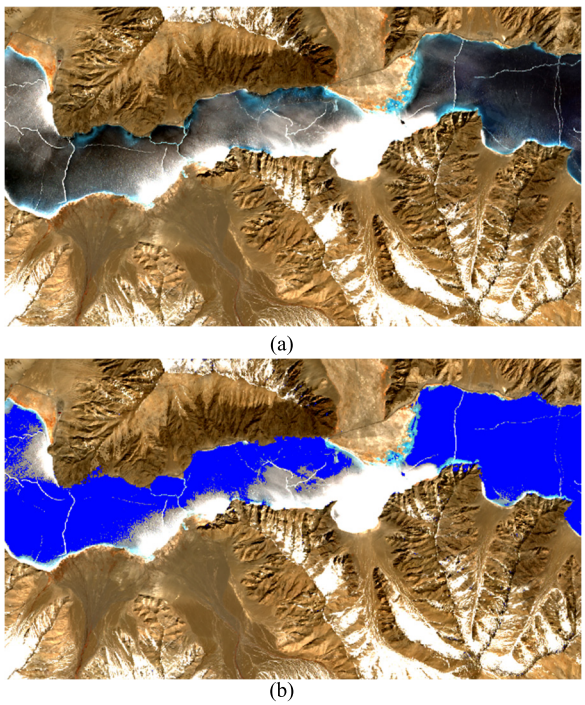

Fig. 4. Water layer extraction in the Himalayan region. (a) Pre-monsoon FCC image. (b) Extracted water layer overlaid on FCC image.

vegetation as the second constituent showed a larger deviation than those with soil/land as the second constituent. This is because the difference in NDWI is more prominent between the water and vegetation classes than between water and soil classes.

Since the abovementioned issue is expected in region-growing techniques, it was initially planned to use a layer where the difference between water/vegetation classes is similar to the water/soil classes' difference. A layer generated by averaging the NDWI and MNDWI-2 seemed to be a promising option. MNDWI-2, due to its stronger sensitivity to water, was

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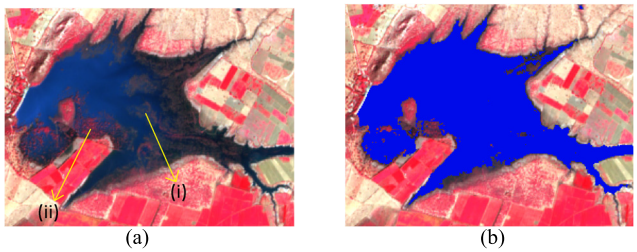

Fig. 5. Water Extraction in (i) sparsely and (ii) densely spaced floating vegetation. (a) Water body with floating vegetation. (b) Extracted water layer overlaid on FCC image.

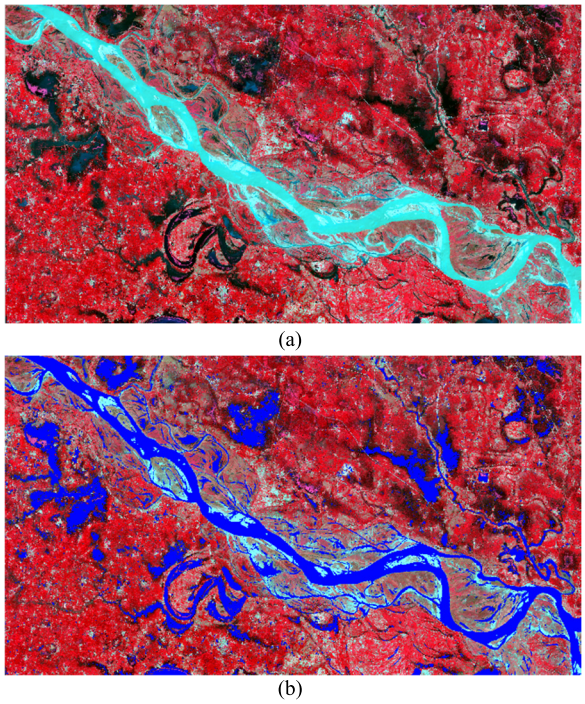

Fig. 6. Standing water extraction in flooded areas. (a) FCC image of standing water after the flood event. (b) Extracted water layer overlaid on FCC image.

expected to reduce the difference between both types of mixed pixels. However, it was observed that the user's accuracy was reduced due to the commission of many nonwater pixels, especially in moist and damp soils. Hence, only the NDWI layer is used in the region growing. Therefore, classification suffered in areas where waterbodies are covered with dense vegetation.

Extraction of standing water after/during floods is another challenging scenario. Many studies [20], [21], [22] generally use a combination of optical and SAR datasets to extract the flood layer. In optical data, multiple indices are generally used to extract flood water [21], [23] due to the mixed pixels with different types of constituents. To understand and analyze the performance of this adaptive region-growing algorithm in these areas, a post flood-event scene (Fig. 6) was also included in the evaluation. An accuracy of 94.4% was achieved showing the applicability of this algorithm in such areas. The highly turbid water in the river, the flooded agricultural fields, and settlements were delineated very well using this technique. Since thresholds are not manually determined in this algorithm, it is less prone to bias and executes faster to generate inundation maps during any emergency.

IV. Conclusion

The technique used in this study is entirely automatic, and no manual intervention is required. The results of this algorithm are satisfactory and can be safely assumed to be the lower bound due to the challenging scenes selected for accuracy assessment. Considering the achieved accuracy levels and the large amount of data to be processed in near real time, this technique can be a good substitute for manual water extraction algorithms.

The technique for determining the threshold is localized as it utilizes the statistics of the previously classified water pixels in the same region. It is also adaptive, as it changes with every new pixel added. This helps to accurately delineate water spread area for water bodies of various shapes, sizes, and conditions. Being a semi-object-based technique, this algorithm requires less computational time than traditional object-based techniques. In addition, the region growing is applied only on a single layer (NDWI). This also reduced the computational time for the algorithm. The algorithm may be improved further by using multiple layers for region growing, and probability-based object formation can also be introduced.

Acknowledgment

The authors would like to thank Director, NRSC, Deputy Director, RSA, and Group Director, WRG, NRSC, for their encouragement and technical guidance during the study. They would like to place on record their sincere gratitude to B. Simhadri Rao, Sai Krishna, and Anurag Mishra, Engineers from WRG, for evaluating the performance of this algorithm and providing feedback. They would also like to thank the anonymous reviewers for their valuable comments, which improved this article significantly.

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